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Genomics Define Malignant Transformation in Myeloma Precursor Conditions

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ABSTRACT

Multiple myeloma (MM) is consistently preceded by monoclonal gammopathy of undetermined significance (MGUS) and smoldering multiple myeloma (SMM). While these precursor conditions are asymptomatic, they are not entirely benign and carry a lifelong risk of progression to MM. Unlike other cancers defined by pathology, malignant transformation from MGUS or SMM to MM has so far relied on demonstration of clinical end-organ damage as morphology and cytogenetics cannot reliably distinguish them. In this study, using genomic data from 374 patients with MGUS or SMM (277 training, 97 validation), to our knowledge, we demonstrate for the first time the ability to identify malignant transformation in MGUS and SMM. We introduce the concept of genomic MM and genomic MGUS to differentiate the subsets of MGUS and SMM that are biologically malignant with genomic features indistinguishable from MM from the subset that is premalignant and unlikely to progress to malignancy. Importantly, we find that most SMM has biological features of malignant transformation indistinguishable from MM. As expected, this subset that we consider having genomic MM is associated with a high risk of progression to MM although some patients remained progression-free beyond 5 years. Conversely, 60% of MGUS and 10% of SMM have no evidence of malignant transformation (genomic MGUS), with no progression during follow-up. Integration of genomic features with the 2/20/20 International Myeloma Working Group model significantly improved the prediction of progression among genomic MM. These findings support the use of genomic criteria to refine the classification and the risk stratification in myeloma precursor conditions.

ACCOMPANYING CONTENT

Data Sharing Statement

Data Supplement

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INTRODUCTION

Multiple myeloma (MM) is consistently preceded by the asymptomatic expansion of clonal plasma cells, termed either monoclonal gammopathy of undetermined significance (MGUS) or smoldering myeloma (SMM).[1-5] These two clinical entities were defined over four decades ago based on the size of the aberrant plasma cell clone, and the classification criteria have not been revised since.[6] Unlike other malignancies, the distinction of premalignancy from malignancy across this spectrum of neoplastic plasma cell disorders is not possible with standard morphologic or cytogenetic studies since they all share the same features. As a result, MGUS and SMM have been traditionally differentiated from MM mainly based on clone size and the presence or absence of clinically detectable end-organ damage. This requirement for end-organ damage has been a barrier for progress in the field in terms of early detection and intervention studies. Furthermore, 3%-5% of the adult population has MGUS or SMM, whereas only a subset will ultimately progress to MM, and more accurate methods of risk stratification are needed.[7-10]

To reduce the need for clinical organ damage as a requirement to diagnose malignant transformation (MM),[11] the International Myeloma Working Group (IMWG) defined a subset of patients with SMM who lack organ damage but show early evidence of emerging disease activity using modern imaging (eg, magnetic resonance imaging [MRI]) and biomarkers (ie, serum free light chains and plasma cell infiltration in the bone marrow) and at imminent risk of progression as MM.[6] This change has allowed clinicians to

CONTEXT

Key Objective

Is it possible to use genomics to predict which patients with myeloma precursor conditions—monoclonal gammopathy of undetermined significance (MGUS) and smoldering multiple myeloma (SMM)—will remain stable or progress to active multiple myeloma (MM)?

Knowledge Generated

Using genomic data from 374 patients with MGUS or SMM, we introduce the concepts of genomic MM and genomic MGUS to reclassify patients into two biologically distinct groups: a malignant subset with a significant risk of progression to active MM and a truly premalignant group with no genomic evidence of malignancy and no risk of progression. By restricting outcome prediction to patients with genomic MM, we show that integrating genomic features with existing clinical risk stratification models significantly enhances the accuracy of predicting progression.

Relevance *(S. Lentzsch)*

This study demonstrates that whole-genome and exome sequencing can define biologically distinct subsets of MGUS and SMM, enabling a genomics-based classification with clinical utility. By focusing predictive modeling on truly malignant precursor states, the findings lay the groundwork for refining diagnostic criteria and guiding early intervention in MM.*

*Relevance section written by *JCO* Associate Editor Suzanne Lentzsch, MD, PhD.

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investigate treatment earlier with the aim of preventing organ damage. Although this and other risk stratification efforts enabled some patients to avoid organ damage and to the development of early intervention strategies for SMM,[12-19] it has become even more important to distinguish patients with MGUS and SMM who although asymptomatic have genomic evidence of biological malignancy and are at higher risk of progression compared with those who have an indolent, premalignant condition and can be reassured.

Randomized clinical trials have demonstrated clear benefits in terms of progression-free survival and, more recently, overall survival in persons considered to have high-risk SMM defined mainly using markers of clone size.[16-18] A key challenge in these trials lies in the enrollment criteria, which primarily focus on high-risk SMM—a definition that has evolved over time, often before the key trials completed enrollment.[20,21] This observation has prompted several research groups to develop prognostic models aimed at identifying patients with SMM at imminent risk of progression who would truly benefit from early intervention.[8,22-24] While the current gold standard, the IMWG 2/20/20 risk model,[22] successfully identifies patients with a high risk of progression, it has limitations. For instance, some high-risk IMWG 2/20/20 patients may not progress for several years, and identification of these patients can prevent overtreatment in individuals without a clear clinical need or immediate risk of organ damage. On the other hand, patients classified as not having high-risk SMM may be at risk of early progression and need to be identified for closer follow-up.

Since MM progression is driven at least partly by the accumulation and selection of distinct genomic alterations, several studies have explored the role of genomic profiling in distinguishing progressive from stable MGUS and SMM.[25-31] Although often limited in sample size and being primarily focused on SMM, these studies have identified specific genomic events linked to progression risk such as *MYC* translocations and RAS mutations.[25-31] This finding underscores the potential of genomic profiling to differentiate clonal entities that have undergone malignant transformation from those that are not independent of traditional markers of disease burden. However, previous studies have consistently analyzed MGUS and SMM as separate entities, and the use of genomics to refine this historical classification has never been explored.

The aim of this study is to leverage comprehensive genomic profiling to redefine myeloma precursor conditions, and this, in turn, will improve risk prediction for progression to MM. To do so, we assembled a large training and validation set of patients with SMM and MGUS with available comprehensive genomic profiling to develop a robust classification model that distinguishes patients with evidence of malignant transformation in the clonal plasma cells from those without. Next, we identified distinct myeloma genomic defining events associated with progression in patients with evidence of malignant transformation to MM. By integrating genomic and clinical data, we show that genomic profiling significantly improves the predictive accuracy of the current gold standard—the IMWG 2/20/20 prognostic model—and helps identify patients whose disease is already progressing.

These findings underscore the critical role of genomic profiling in redefining myeloma precursor conditions, with important implications for future translational research in this field.

METHODS

A total of 374 patients with SMM (n = 290) or MGUS (n = 84) with available whole-exome sequencing (WES, n = 190) or whole-genome sequencing (WGS, n = 184) were included in the study (Data Supplement, Tables S1 and S2, online only). All sequencing FASTQ files were aligned to Genome Reference 38, centralized, and analyzed using the Myeloma Genome Project pipeline (Data Supplement). All tumor samples were sorted using either CD138+ magnetic beads or flow cytometry. All tumor samples were analyzed using as germline match normal, either peripheral blood mononuclear cells or CD138-negative bone marrow cells. Careful genomic quality control was performed in each patient by integrating copy number variants (CNVs), single-nucleotide variants, the presence of oncogenic rearrangement, and/or productive V(D)J in the immunoglobulin regions to confirm the presence of a clonal B-cell population and assess for potential contamination of normal cells into the tumor samples. The analytical workflows used for this study are available in the Data Supplement (Fig S1) and in github.

RESULTS

Study Population

Among the 374 patients with high-quality sequencing data, 62 (16.5%) were enrolled in SMM early intervention clinical trials.[12,15] An additional 11 (3%) did not have complete follow-up. These patients were only included in the classification part of the study and not included in the time-to-event analysis (Data Supplement, Fig S1). A total of 58% and 78% of patients with MGUS and SMM, respectively, had a radiologic evaluation to exclude bone lesions at the time of the bone marrow biopsy with X-ray, positron emission tomography-computed tomography, or MRI. Overall, none of the patients had evidence of SLiM or CRAB criteria at the time of the sampling.[6]

A total of 301 patients (81 MGUS and 220 SMM) were followed over time and treated only in the case of organ damage or SLiM criteria.[6,11] Disease progression was defined in accordance with IMWG SLiM-CRAB diagnostic criteria for MM.[6] Time to progression into MM was defined from the time of sample collection to the last follow-up or progression. As expected, patients with SMM experienced more and earlier progressions to MM than MGUS ($P < .0001$; Fig 1A). Overall, after a median follow-up of 46 months, 80 of 220 (36%) patients with SMM and 4 of 81 (5%) patients with MGUS experienced progression ($P < .0001$). The key clinical features for patients with MGUS and SMM are summarized in the Data Supplement (Table S1).

Based on the proportion of SMM and MGUS patients with WES or WGS, all eligible patients from Mayo Clinic and the MDACC were included as the validation set (n = 97). The remaining patients were assigned to the training set (n = 277, 68 of which were either lost to follow-up or enrolled on an interventional clinical trial). In the training and validation set, data to calculate risk of progression into MM from SMM according to the IMWG 2/20/20 model were available for 161 (76.5%) and 74 (76.2%) patients, respectively. In both data sets, IMWG 2/20/20 was able to identify SMM patients with significantly shorter time to progression (Figs 1B and 1C). However, differences in clinical outcomes between IMWG 2/20/20 low- and intermediate-risk groups were observed only in the validation set. Overall, the IMWG 2/20/20 c-index to predict SMM progression in the training and validation set was 0.67 (0.63-0.71) and 0.72 (0.68-0.76), respectively (Fig 1D). Recently proposed clinical markers for predicting SMM progression, such as hemoglobin and creatinine, did not show any independent prognostic value in predicting the risk of progression in either the training or validation cohort (Fig 1E).[24]

Genomic Profiling to Define Malignant Transformation

By integrating CNV and single base substitution (SBS)/indel data from 1,933 newly diagnosed MM (NDMM) patients, a catalog of 134 myeloma genomic defining events was recently proposed.[34] Leveraging this model, we investigated the prevalence and distribution of these 134 events among SMM and MGUS patients and their association with progression into MM in both the training and validation cohorts (Data Supplement, Fig S2). Given the distinct clinical setting, a few modifications were implemented from the original model. Specifically, in the setting of NDMM, CNV signatures were previously used to predict for the prevalence of chromothripsis.[35,36] However, because chromothripsis is infrequent, between SMM and MGUS, we decided to reanalyze de novo CNV signatures, identifying five distinct CNV-signatures (Data Supplement, Fig S3). Three signatures were characterized by multiple aneuploidies—including focal events—and enriched in SMM versus MGUS and in patients who progressed (SMM CNV SIG 1, 3, and 5). The remaining signatures were marked by minimal or absent CNVs and was associated with MGUS and SMM patients that remained stable for decades (SMM CNV SIG 2 and 5).

Regarding mutational signatures, in NDMM, most patients show evidence of aberrant apolipoprotein B mRNA editing enzyme (APOBEC) activity and the focus is typically on identifying those with higher levels as this is prognostic (ie, hyper-APOBEC).[34,37] In the context of myeloma precursor conditions, we instead focused on the APOBEC presence versus absence, given that APOBEC-mediated mutagenesis is a known driver of malignant transformation in solid tumors.[38-42] To achieve this, APOBEC SBS signatures (SBS2 and SBS13) were assessed only in patients with an overall mutational burden >50 SBS (Data Supplement, Fig S4).[43]

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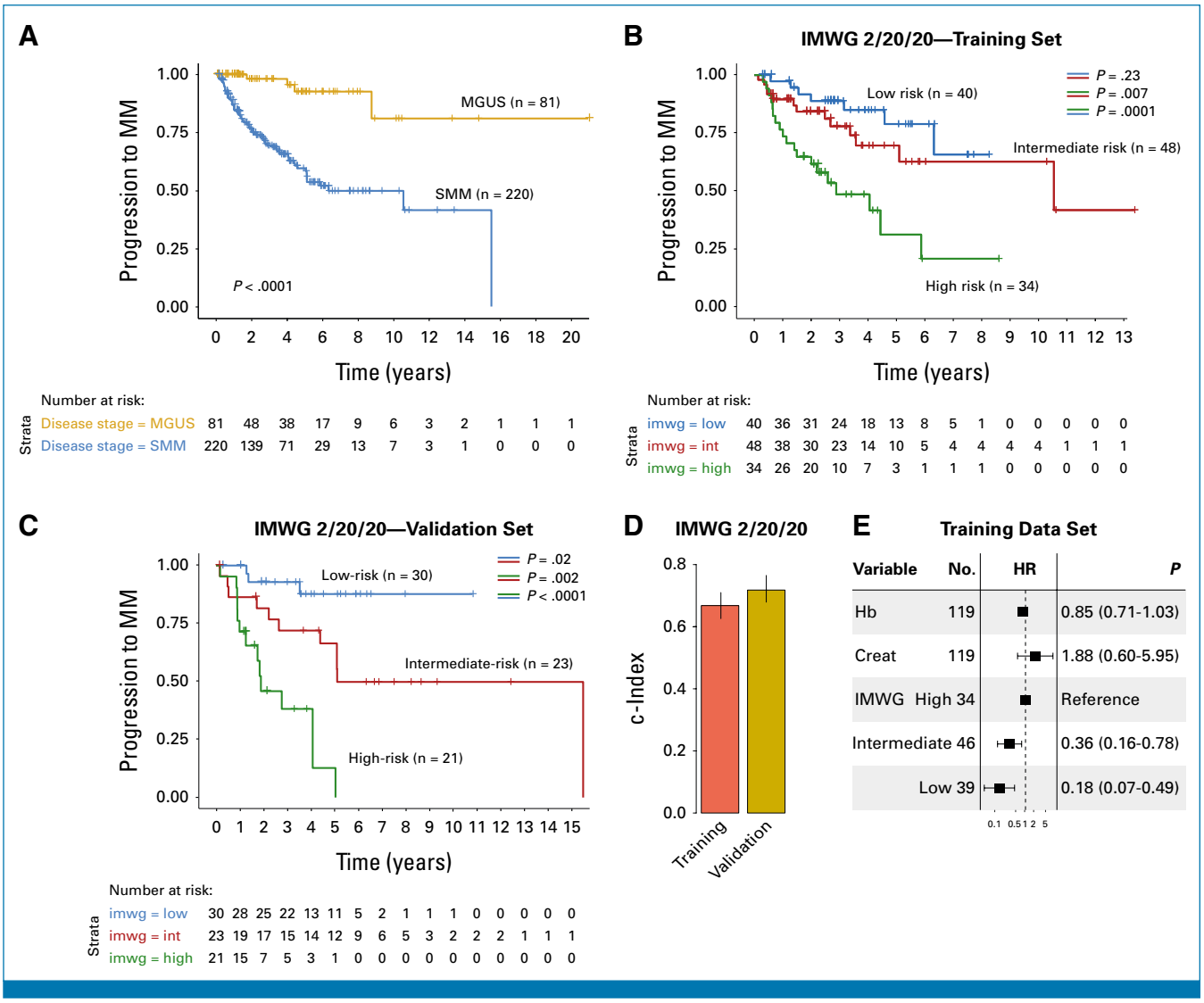

FIG 1. Clinical predictors of progression in myeloma precursor conditions. (A) Kaplan-Meier curve comparing risk of progression between MGUS and SMM. Prognostic performance of the 2020 IMWG model in the (B) training and (C) validation cohorts. (D) Concordance index (c-index) of the IMWG 2020 model in training and validation sets. (E) Multivariate Cox proportional hazards model evaluating the association between baseline clinical parameters and risk of SMM progression in the training set. Included covariates are serum creatinine (ie, Creat), Hb, and the IMWG 2020 risk stratification. Among patients presenting with high creatinine or anemia, these abnormalities were always attributable to pre-existing conditions and not to MM or myeloma precursor conditions. Hb, hemoglobin; HR, high risk; IMWG, International Myeloma Working Group; MGUS, monoclonal gammopathy of undetermined significance; MM, multiple myeloma; SMM, smoldering multiple myeloma.

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Integrating the above data, a total of 28 myeloma genomic defining events were identified to be associated with progression to MM in the training set either considering all patients and limited to SMM (Fig 2; Data Supplement, Tables S3, S4, and S5). This list included the presence of APOBEC mutagenesis, t(4;14) (*NSD2*;*IGH*), *RAS* mutations, hyperdiploidy (HY), and *MYC* translocations. Interestingly, we observed a group of patients in both the training and validation sets who did not exhibit any of these events and were characterized by an extremely indolent genomic profile (Fig 2). The mutational and CNV profiles of these patients were remarkably similar to those observed in normal B cells,[42] suggesting that the malignant transformation has not yet occurred in these patients—and may never occur. By contrast, the other patients exhibited varying degrees of genomic complexity and aberrations consistent with myeloma genomic defining events.[34] In fact, the genomic profile of most of these patients in both the training and validation sets closely resembled, or was identical to, the profiles observed across more than 2,000 patients with NDMM.[34] This suggests that this group includes patients in whom malignant transformation to MM has already occurred at the genomic level, although clonal expansion leading to organ damage has not yet taken place. Based on these data and 28 myeloma genomic defining events associated with progression, we used the training set to develop a workflow that differentiates MGUS and SMM into two genomically distinct entities: one with evidence of malignant transformation (ie,

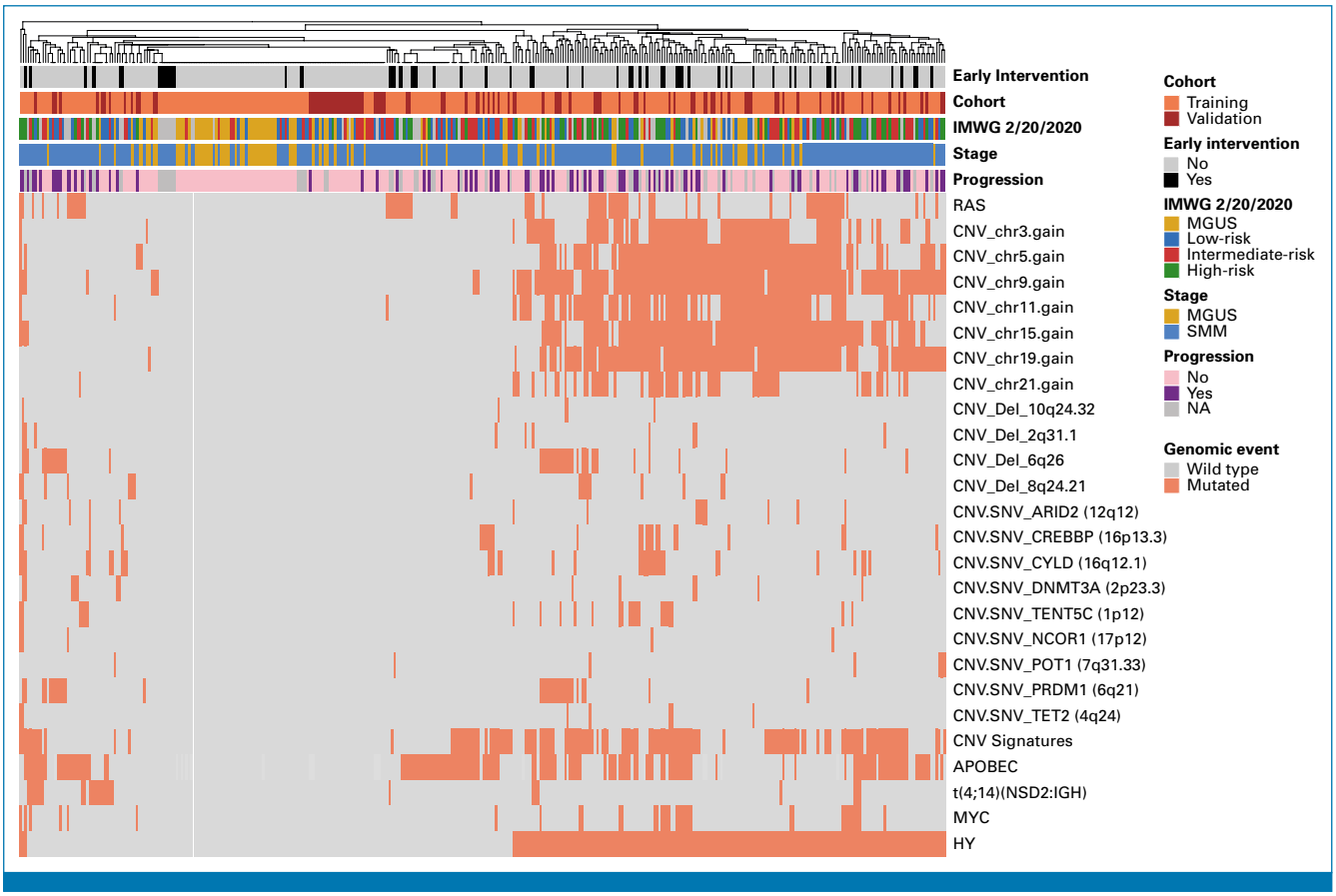

FIG 2. Heatmap summarizing the genomic features associated with myeloma precursor condition progression into MM. Events involving RAS were grouped together in line with previous evidences.[27,31] Genomic events are reported using the annotation used by Maura et al.[34] IMWG, International Myeloma Working Group; MGUS, monoclonal gammopathy of undetermined significance; MM, multiple myeloma; NA, not available; SMM, smoldering multiple myeloma.

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genomic MM) and one without (ie, genomic MGUS; Fig 3A; Data Supplement, Figs S5 and S6 and Tables S6 and S7). Overall, 39% of MGUS and 94% of SMM patients had genomic evidence of malignant transformation in the training set (ie, genomic MM). Importantly, none of the patients with genomic MGUS experienced progression into MM in the training set (Fig 3B). When we applied the same workflow and classification to the validation set, 47% of MGUS and 85% of SMM had evidence of malignant transformation. Importantly, also in the validation set, none of the genomic MGUS experienced progression (Fig 3C). All patients with IMWG 2/20/20 high-risk SMM were classified as genomic MM in both the training and validation set (Fig 3A). Interestingly, 5 of 62 (8%) patients enrolled in early intervention trials for high-risk SMM were genomically classified as genomic MGUS. Overall, these data demonstrated that the MM genomic background and malignant transformation can be acquired very early in time, even before the clinical SMM phase as defined currently[6] (eg, Figs 3D-3F).

Myeloma Genomic Defining Events Associated With SMM Progression

Because of the comprehensive genomic resolution and the workflow described above, we identified patients in whom malignant transformation had already occurred (ie, genomic MM), despite clinical parameters indicating early and asymptomatic disease. By contrast, the other patients (ie, genomic MGUS) neither experienced clinical progression—regardless of their SMM or MGUS status—nor exhibited any evidence of malignant transformation. Therefore, it is essential to exclude the latter patients from any prognostic modeling as their inclusion would distort and overfit the prognostic impact of different myeloma genomic defining events, which are, by definition, enriched in the genomic MM group compared with the genomic MGUS group. In fact, distinct genomic drivers may predict malignant transformation but not necessarily the risk of progression to MM once transformation has occurred. Among the patients with genomic MM in the training set who were not enrolled in early intervention clinical trials (n = 158), 55 (35%) experienced progression and 29 (17%) remained stable for more than 5 years, despite having a myeloma-transformed genomic profile (Figs 3B-3F).

To identify the genomic features underlying this clinical heterogeneity, we investigated the prognostic impact of each myeloma genomic defining event in predicting MM progression in the training set. After running 5-fold cross-validations corrected for IMWG 2/20/20, only the

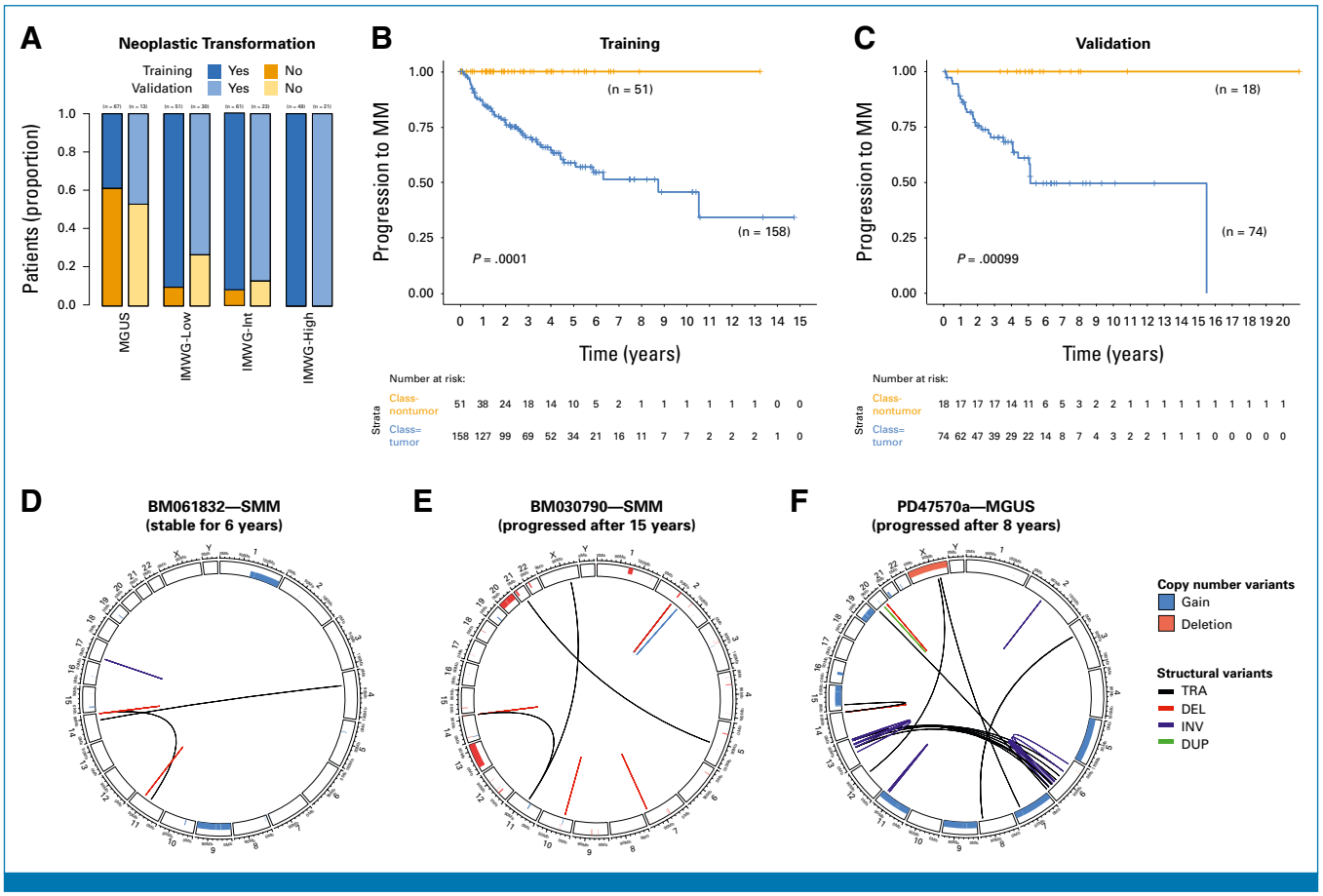

FIG 3. Malignant transformation in myeloma precursor conditions. (A) Proportion of patients with MGUS and SMM, stratified by IMWG 2020 risk criteria, showing evidence of malignant transformation (ie, genomic MM). Clinical impact of malignant transformation in the (B) training and (C) validation cohorts. (D-F) Circos plots illustrating three representative cases of patients with genomic evidence of malignant transformation who remained clinically stable for several years. (F) Genomic events include aneuploidies, chromosomal gains, and complex structural variants such as chromothripsis. DEL, deletion; DUP, duplication; IMWG, International Myeloma Working Group; INV, inversion; MGUS, monoclonal gammopathy of undetermined significance; MM, multiple myeloma; SMM, smoldering multiple myeloma; TRA, translocation.

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presence of *RAS* mutations, genomic events involving *MYC*, the presence of neoplastic CNV signatures, and APOBEC mutagenesis were predictive of early progression in at least two cross-validations (Figs 4A-4D; Data Supplement, Fig S7). Patients with at least two of these four genomic events had rapid progression in both the training and the validation set (Data Supplement, Fig S8). Adding the presence of high-risk myeloma genomic defining events into the IMWG 2/20/20 (ie, genomic IMWG; Data Supplement, Figs S9 and S10; Fig 5A) significantly improves the accuracy in predicting MM progression in both the training (c-index 0.69 *v* 0.74) and validation set (c-index 0.74 *v* 0.79; Figs 5B-5E). Using the genomic IMWG model developed in the training set to create a weighted risk score by estimated log high risk in the validation set showed a c-index of 0.77 (0.73-0.81; Data Supplement). Finally, restricting the analysis to the patients with radiologic assessment at the time of the biopsy confirmed the accuracy improvement of the genomic IMWG model (Data Supplement, Fig S11). Overall, these data show that comprehensive genomic profiling can improve the accuracy of predicting progression to MM in SMM patients with genomic MM.

DISCUSSION

Unlike other malignancies, it is not possible using standard histopathology to differentiate malignancy from premalignancy in clonal plasma cell disorders, necessitating the use of crude tumor burden and clinical parameters such as end-organ damage to define MM, and to distinguish it from MGUS and SMM.[6] Yet, given the high prevalence of plasma cell dyscrasias in the general population[10] and the benefit of early intervention in high-risk SMM,[16] there is a compelling need to redefine the risk stratification in such asymptomatic conditions.

To our knowledge, our study shows for the first time that 90% of SMM, including all patients with high-risk SMM, has malignancy signatures indistinguishable from MM.[34] By introducing the concept of distinguishing genomic MGUS from genomic MM, we propose a new genomic-based

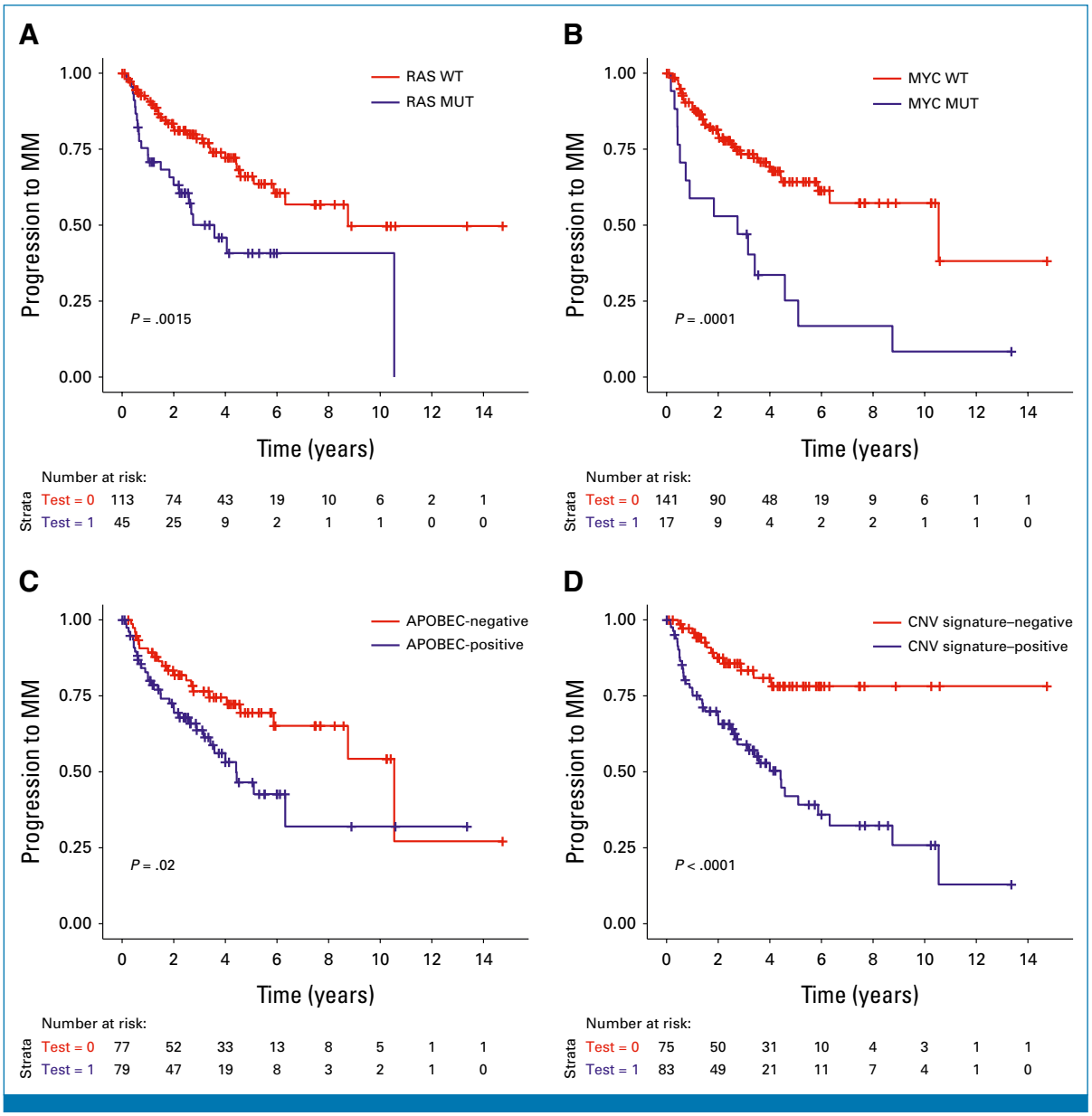

FIG 4. Myeloma genomic defining events predictors of progression in myeloma precursor conditions with evidence of malignant transformation (ie, genomic MM) in the training set. Clinical impact of specific genomic features: (A) RAS mutations, (B) genomic events involving *MYC* and CNV, (C) signatures indicative of genomic complexity, and (D) APOBEC mutagenesis, defined as presence or absence, on outcomes in patients with myeloma precursor conditions with evidence of malignant transformation. APOBEC, apolipoprotein B mRNA editing enzyme, catalytic polypeptide-like; CNV, copy number variant; MM, multiple myeloma; WT, wild type.

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classification model that significantly improves the one developed by Kyle et al 3 decades ago.[2,8] In solid tumors, it is well known that in situ carcinomas and cancerous cells, such as those found and characterized in Barrett esophagus, exhibit multiple genomic alterations commonly found in cancer, despite remaining quiescent for years.[44,45] Through various orthogonal approaches, we now understand that most tumors, including MM, develop over decades, with the early acquisition of genomic events that likely initiate neoplastic transformation.[9,46-49] Afterward, the tumor expands following classical Darwinian evolution, with patterns of neutrality and selection.[4,50,51] Our study shows that this pattern is present in clonal plasma cell disorders and that evidence of malignant transformation can be detected in most SMM patients and a fraction of MGUS. This finding confirms our hypothesis based on the well-established clinical course of these disorders and the fact that the absence of symptoms or end-organ damage does not mean that these are entirely benign or premalignant states.

Our findings have major clinical implications beyond the biological distinction of premalignancy and malignancy. The

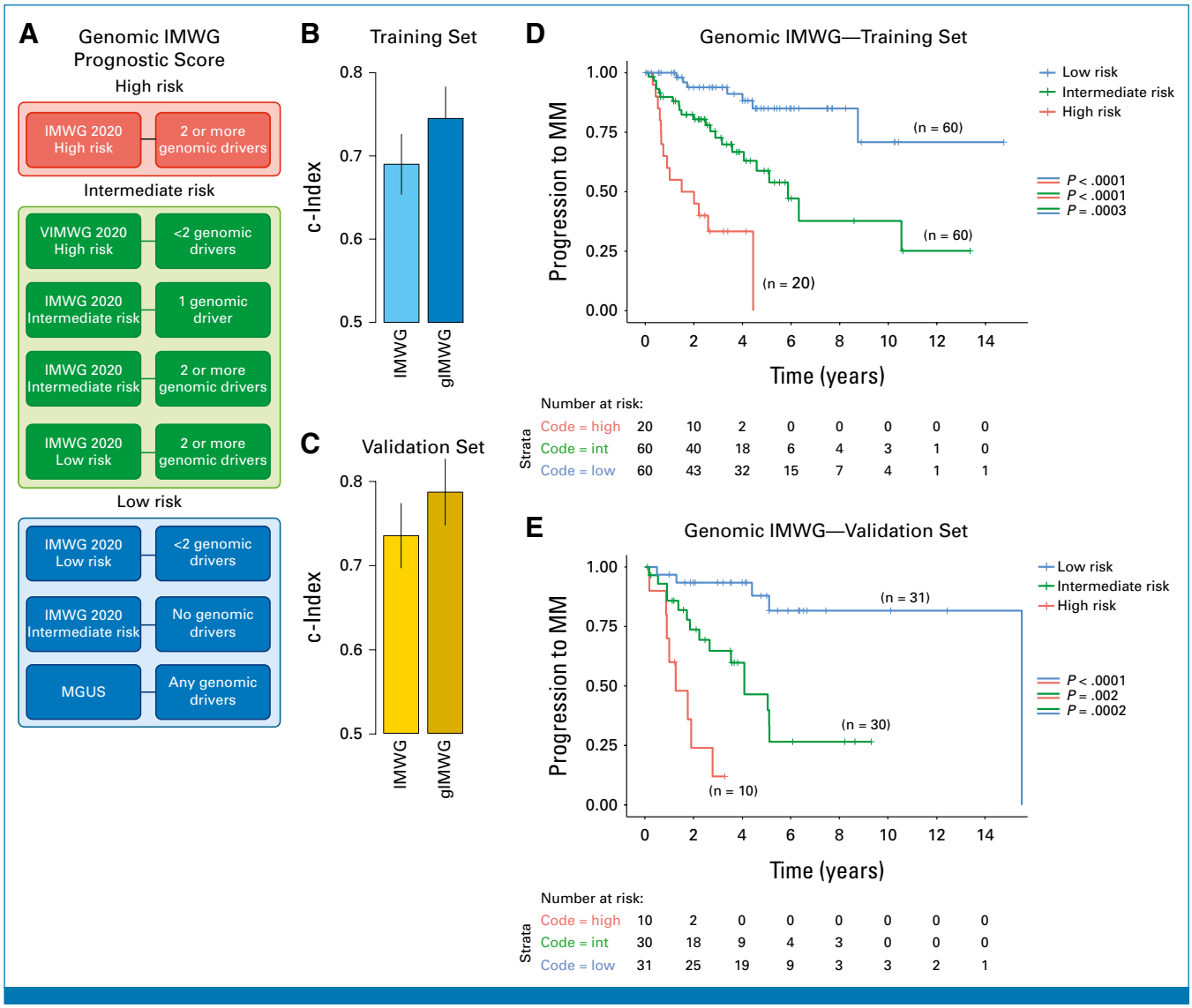

FIG 5. Genomic IMWG prognostic score. (A) Schema of how IMWG 2020 and the genomic score proposed in this study were integrated. Improvement in the concordance index (c-index) upon integrating genomic features with the IMWG 2020 model in the (B) training and (C) validation data sets. Kaplan-Meier curves illustrating progression to MM stratified by risk groups defined through the combined IMWG 2020 criteria and selected genomic features in the (D) training and (E) validation sets. IMWG, International Myeloma Working Group; MM, multiple myeloma.

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identification of patients with SMM and MGUS who will progress and experience organ damage has been a key priority in the MM community for decades. With the advent of novel and highly effective immunotherapies that have shown clear advantages in SMM,[16,17] the need for more accurate models has become apparent—to better identify patients at imminent risk while preventing unnecessary exposure to potential side effects in those with indolent disease. Our study leads to an important shift in our understanding, defining genomic MM as the equivalent of malignancy but without evidence of end-organ damage that has traditionally been a requirement to define MM in clinical practice, the equivalent of stage I-II chronic lymphocytic leukemia or asymptomatic low-grade follicular lymphoma.[52,53] This is of particular importance for SMM where we find that 90% has evidence of malignant transformation and that our proposed genomic markers and classification can further refine the risk stratification of SMM and enable us to identify with greater accuracy candidates for early intervention. An important caveat is that although 40% of the MGUS patients in our study showed evidence of malignant transformation and would be considered genomic MM, this is likely a significant overestimation as bone marrow biopsies are not recommended in low-risk MGUS and since patients with MGUS followed in large cancer centers—such as those involved in this study—tend to have a higher disease burden and risk compared with MGUS diagnosed in the community among the general population.[54,55] Furthermore, this high proportion can also be explained by the exclusion of MGUS with

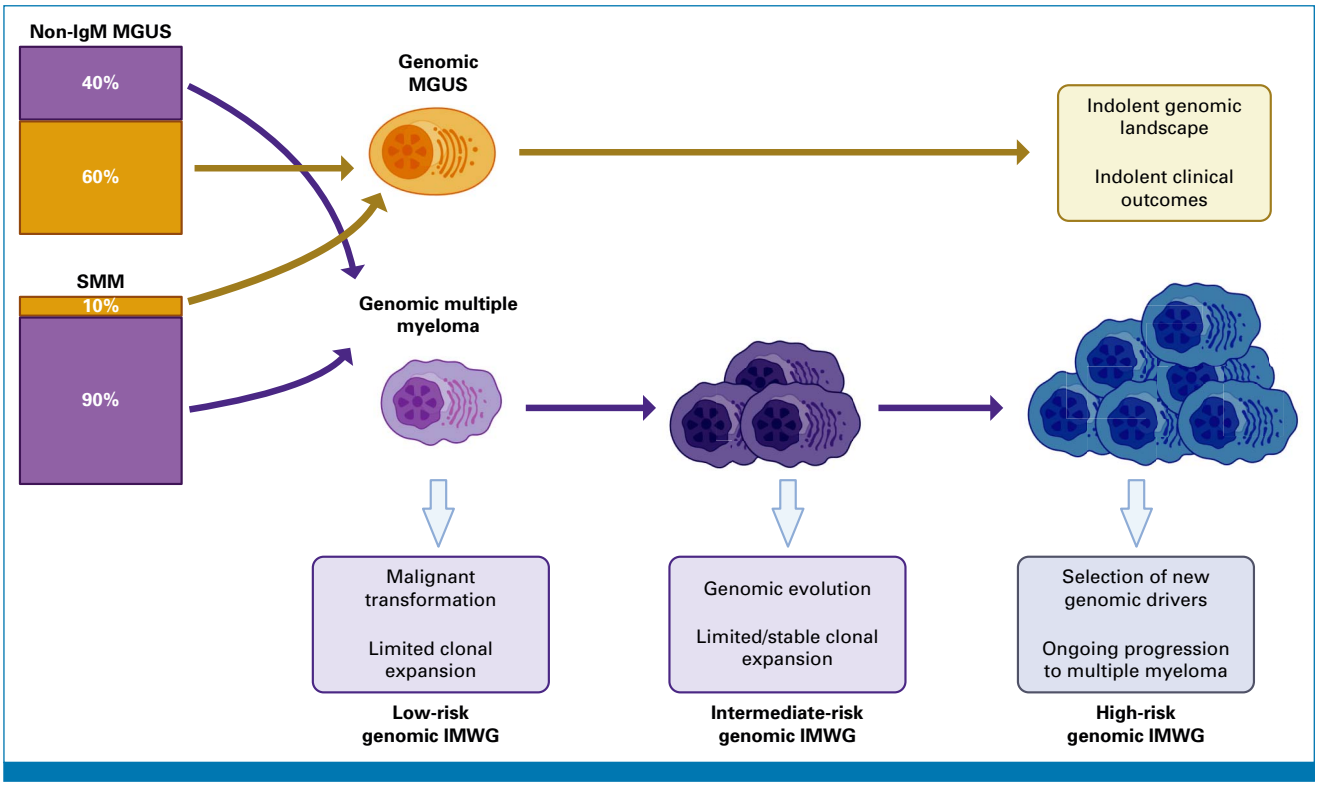

FIG 6. Cartoon summarizing the new classification and prognostic model proposed in this study. MGUS includes only patients with evidence of clonal plasma cell in the bone marrow assessed by whole-genome or whole-exome sequencing. IgM, immunoglobulin M; IMWG, International Myeloma Working Group; MGUS, monoclonal gammopathy of undetermined significance; SMM, smoldering multiple myeloma.

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significantly more polyclonal than clonal bone marrow plasma cells, whose genomics is not accessible by bulk sequencing of CD138-selected cells because of the lack of the dominant clonal population. Importantly, because of the resolution limitations of bulk sequencing, we cannot rule out the presence of minor subclones harboring transformation-associated alterations that remain undetectable at the time of sampling or are located in anatomic sites not captured by standard bone marrow biopsies. While the 5-year risk of progression in genomic MGUS is zero in both the training and validation set, the possibility of transformation over longer periods cannot be entirely excluded—especially considering that MM may take 30-40 years to fully develop.[48,49] This underscores the importance of integrating genomic and clinical variables to guide appropriate follow-up strategies for patients with myeloma precursor conditions.

Interestingly, some patients with genomic MM remain stable for years despite having a genomic profile similar to NDMM (Figs 3D-3F). This behavior could be explained by extrinsic factors such as immune control, whose dysregulation over time because of intrinsic or extrinsic factors might let the MM clone expand and cause organ damage. Evidence for immune-mediated control has previously been demonstrated in xenograft models, where human MGUS-derived cells exhibited progressive growth in mice—suggesting that the clinical stability of these lesions may partly result from growth restraints imposed by the immune system rather than intrinsic tumor cell properties.[56] In contrast to premalignant solid tumors or carcinoma in situ, which may be excised, MM-transformed clones are not easily eradicated, even with intense therapies.[12,21,57] Therefore, strategies aimed at maximizing the pre-existing immune control to keep these neoplastic clones in check and limiting the selection of more aggressive clones are attractive and have already shown some advantages[16,17] although further genomic and integrated immune studies are needed. Other early intervention strategies with more aggressive approaches might be more effective on killing the aberrant plasma cell, but they might also irreversibly affect this immune control.

Importantly, the genomic classification proposed here is not primarily designed for prognostication, but rather to distinguish two biologically distinct entities—one of which follows an indolent and likely benign clinical course. Including these patients, who typically lack myeloma genomic defining events and never progress, in analyses with genomic MM may dilute associations or lead to overfitting in prognostic models built on key genomic features. Therefore, to accurately assess the risk of progression in myeloma precursor conditions, we should focus solely on the biological subset that is at genuine risk of progression: genomic MM (Figs 5-6). In fact, while some of these patients remain stable for years, others exhibit genomic profiles that not only

resemble those of MM but also suggest a high risk of imminent or ongoing progression. In line with previous evidence,[25,27,29-31,58] our data suggest that these alterations associated with rapid progression reflect activation of *MYC* and RAS pathways and genomic instability (ie, CNV signatures and APOBEC mutagenesis). It is currently unclear which approach—myeloma-like treatment or immune control—would be most effective for these patients, and multiple ongoing clinical trials might provide soon the answer. In the recent AQUILA randomized phase III clinical trial, 20% of patients with SMM progressed within a few months of initiating daratumumab treatment.[16,59] This subgroup likely represents patients in whom immune surveillance mechanisms are ineffective, potentially corresponding to the high-risk group of genomic MM reported here. Confirming these findings in future studies will be essential as these patients may require early initiation of multiagent antimyeloma therapy.

While the results of different trials will be released in the coming years, it is important to highlight that combining genomic studies in SMM and MGUS with traditional IMWG 2/20/20 risk stratification will be critical for the interpretation of clinical trial results. Since almost all patients with SMM (90%) have genomic MM, genomics alone is insufficient to identify patients at imminent risk of progression who may benefit from early intervention. On the other hand, since there are some patients with SMM (and the majority with MGUS) who do not have genomic MM, we risk overtreating patients who might have genomic MGUS. Therefore, we recommend combining genomic studies to distinguish genomic MM versus genomic MGUS along with IMWG 2/20/20 (ie, genomic IMWG 2/20/20) or other clinical risk stratification models to identify patients for clinical trials and patients who will benefit the most from early intervention strategies. As all IMWG 2/20/20 high-risk patients were genomic MM in both the training and validation sets, we believe that, despite its limitations, this model should be use as enrollment criteria when more comprehensive genomic profiling strategies are not available, thus providing an accessible and reproducible way to select patients and interpret trials. Furthermore, dynamic models that track changes in disease burden over time are particularly promising[24] as these markers are easily accessible through peripheral blood and highly sensitive in predicting ongoing progression to MM. Unfortunately, such data were not available for this study, making integration of dynamic changes with genomics unfeasible at this time.

While our study did not include longitudinal samples from the same individuals, previous research has demonstrated that SMM can progress to MM through either branching or linear evolutionary trajectories. In both models, the majority of genomic driver mutations are shared between the precursor and transformed stages,[25-27,60] supporting the concept of early transformation as a fundamental feature to identify the patients at risk of progression.

In conclusion, by combining clinical and genomic data from patients with SMM and MGUS, to our knowledge, we leveraged comprehensive genomic profiling to provide for the first time a genomic classification of MGUS and SMM (Fig 6). Based on our study, over 40% of MGUS and approximately 10% of SMM do not have evidence of neoplastic transformation (ie, genomic MGUS), with a need for surveillance rather than intervention. On the other hand, 90% of SMM has genomic malignancy (genomic MM), and within this group, progression to end-organ damage or lack thereof may be dependent to some extent on tumor burden (as estimated by models such as IMWG 2/20/20), immune control, and occurrence and accumulation of other high-risk genomic events. The rapid decrease in sequencing cost and the integration of genomics into clinical practice will offer unprecedented opportunities for clinicians to better assess patients' diagnosis and risk of progression, ultimately improving patients' quality of life and clinical outcomes.

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AUTHORS' DISCLOSURES OF POTENTIAL CONFLICTS OF INTEREST

Genomics Define Malignant Transformation in Myeloma Precursor Conditions

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Honoraria: Celgene, Bristol Myers Squibb, Bayer, Amgen, Janssen, Kite, a Gilead company, Merck Sharp & Dohme, Juno Therapeutics, Takeda, Abbvie, Aduro Biotech, Sanofi, GlaxoSmithKline, Pfizer, Regeneron, RA Capital
Consulting or Advisory Role: Celgene, Bristol Myers Squibb, Bayer, Amgen, Janssen, Abbvie, Kite, a Gilead company, Merck Sharp & Dohme, Juno Therapeutics, Takeda, Aduro Biotech, Sanofi, OncoTracker, GlaxoSmithKline, Karyopharm Therapeutics, Oncopeptides, Pfizer, Pharmacyclics, Regeneron, RA Capital, Adaptive Biotechnologies
Patents, Royalties, Other Intellectual Property: Dr Fonseca has received a patent for the prognostication of MM based on genetic categorization of the disease
Travel, Accommodations, Expenses: Celgene, Bristol Myers Squibb, Bayer, Amgen, Janssen, Kite, a Gilead company, Merck Sharp & Dohme, Juno Therapeutics, Takeda, Abbvie, Aduro Biotech, Sanofi, OncoTracker
Other Relationship: Adaptive Biotechnologies, Caris Life Sciences

Ola Landgren
Honoraria: Celgene, Bristol Myers Squibb, Medscape, Amgen, Janssen, Karyopharm Therapeutics, GlaxoSmithKline
Consulting or Advisory Role: Bristol Myers Squibb, Celgene, Janssen, GlaxoSmithKline, Pfizer
Research Funding: Amgen (Inst), Janssen (Inst), Pfizer (Inst), Menarini Silicon Biosystems (Inst), Sebia (Inst)

Robert Z. Orlowski
Stock and Other Ownership Interests: Asylia Therapeutics, Inc
Consulting or Advisory Role: Bristol Myers Squibb, Janssen, Sanofi, Takeda, Regeneron, GlaxoSmithKline, BioTheryX, Meridian Therapeutics, Monte Rosa Therapeutics, IASO Biotherapeutics, Lytica Therapeutics
Research Funding: Heidelberg Pharma (Inst), Asylia Therapeutics, Inc (Inst), CARsgen Therapeutics (Inst), Sanofi/Aventis (Inst), Exelixis (Inst), Takeda (Inst), Janssen Biotech (Inst), Celgene/Bristol Myers Squibb (Inst)
Patents, Royalties, Other Intellectual Property: Asylia Therapeutics, Inc

Kenneth H. Shain
Honoraria: Bristol Myers Squibb (Celgene), Janssen Oncology, Amgen, GlaxoSmithKline, Sanofi, Abbvie, Karyopharm Therapeutics, Sebia
Research Funding: Pfizer, Abbvie, Karyopharm Therapeutics

Niccolo Bolli
Honoraria: Janssen, GlaxoSmithKline, Pfizer, Jazz Pharmaceuticals, Amgen, Sanofi, Menarini, Oncopeptides
Consulting or Advisory Role: Pfizer, Takeda, Janssen, GlaxoSmithKline
Research Funding: Janssen
Travel, Accommodations, Expenses: Janssen

Saad Usmani
Consulting or Advisory Role: Janssen Oncology, GlaxoSmithKline, Abbvie, Genentech, Bristol Myers Squibb/Celgene, Regeneron, Oricell Therapeutics, AstraZeneca
Research Funding: Janssen Oncology, Bristol Myers Squibb, GlaxoSmithKline, Gracell Therapeutics, Gilead Sciences, K36 Therapeutics

S. Vincent Rajkumar
Honoraria: Research to Practice, Medscape
Patents, Royalties, Other Intellectual Property: Authorship Royalties from Up To Date

No other potential conflicts of interest were reported.

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